



CIS 635 - Knowledge Discovery & Data Mining

Introduction to Data Mining

What is Data Mining?

- **Process of extracting and discovering patterns in large datasets**
- Involves methods: ML, Statistics, DBMS
- Interdisciplinary field: CS, Statistics
- Overall goal:
 - Extracting information from dataset
 - Transform into a comprehensive structure for further use
- Data mining is the analysis step of
 - The KDD
 - Aside from raw analysis, it also involves
 - Database and data management aspects
 - Data preprocessing
 - Modeling and inference considerations
 - Evaluation and metrics
 - Post processing of discovered structures and visualizations

[Wikipedia](#)



What is Data Mining?

- Process of extracting and discovering patterns in large datasets
- **Involves methods: ML, Statistics, DBMS**
- Interdisciplinary field: CS, Statistics
- Overall goal:
 - Extracting information from dataset
 - Transform into a comprehensive structure for further use
- Data mining is the analysis step of
 - The KDD
 - Aside from raw analysis, it also involves
 - Database and data management aspects
 - Data preprocessing
 - Modeling and inference considerations
 - Evaluation and metrics
 - Post processing of discovered structures and visualizations

[Wikipedia](#)



What is Data Mining?

- Process of extracting and discovering patterns in large datasets
- Involves methods: ML, Statistics, DBMS
- **Interdisciplinary field: CS, Statistics**
- Overall goal:
 - Extracting information from dataset
 - Transform into a comprehensive structure for further use
- Data mining is the analysis step of
 - The KDD
 - Aside from raw analysis, it also involves
 - Database and data management aspects
 - Data preprocessing
 - Modeling and inference considerations
 - Evaluation and metrics
 - Post processing of discovered structures and visualizations

[Wikipedia](#)



What is Data Mining?

- Process of extracting and discovering patterns in large datasets
- Involves methods: ML, Statistics, DBMS
- Interdisciplinary field: CS, Statistics
- **Overall goal:**
 - **Extracting information from dataset**
 - **Transform into a comprehensive structure for further use**
- Data mining is the analysis step of
 - The KDD
 - Aside from raw analysis, it also involves
 - Database and data management aspects
 - Data preprocessing
 - Modeling and inference considerations
 - Evaluation and metrics
 - Post processing of discovered structures and visualizations

[Wikipedia](#)



What is Data Mining?

- Process of extracting and discovering patterns in large datasets
- Involves methods: ML, Statistics, DBMS
- Interdisciplinary field: CS, Statistics
- Overall goal:
 - Extracting information from dataset
 - Transform into a comprehensive structure for further use
- **Data mining is the analysis step of**
 - The KDD
 - Aside from raw analysis, it also involves
 - Database and data management aspects
 - Data preprocessing
 - Modeling and inference considerations
 - Evaluation and metrics
 - Post processing of discovered structures and visualizations

[Wikipedia](https://en.wikipedia.org/wiki/Data_mining)





Knowledge Discovery Process (KDP) Models





Knowledge Discovery Process (KDP) Models

Academic Research Models

- Introduced in the mid 1990s
- Several models available
- Suggested steps are similar



Knowledge Discovery Process (KDP) Models

Academic Research Models

- Introduced in the mid 1990s
- Several models available
- Suggested steps are similar

The most widely cited formulation of the KDD process comes from Fayyad, Piatetsky-Shapiro, and Smyth (1996)



Knowledge Discovery Process (KDP) Models

Academic Research Models

- Introduced in the mid 1990s
- Several models available
- Suggested steps are similar

The most widely cited formulation of the KDD process comes from Fayyad, Piatetsky-Shapiro, and Smyth (1996)

5 steps: Fayyad et al KDP model:

- 1) *Selection*
- 2) *Preprocessing*
- 3) *Transformation*
- 4) *Data mining*
- 5) *Interpretation & Evaluation*



Knowledge Discovery Process (KDP) Models

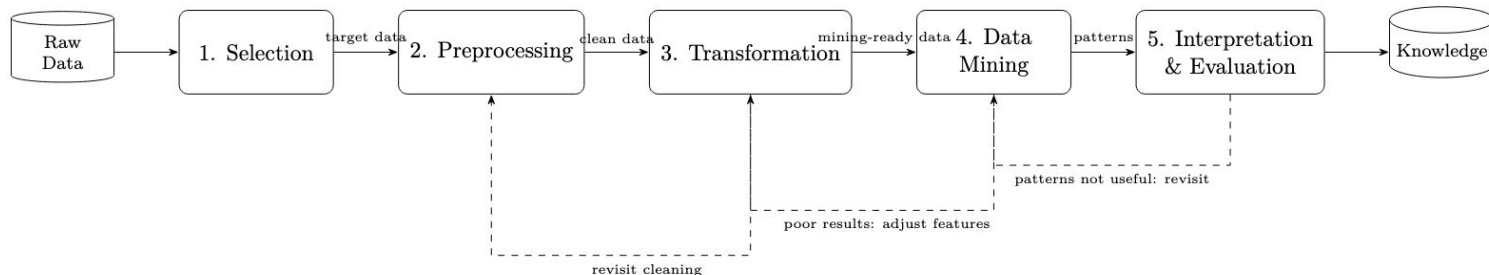
Academic Research Models

- Introduced in the mid 1990s
- Several models available
- Suggested steps are similar

The most widely cited formulation of the KDD process comes from Fayyad, Piatetsky-Shapiro, and Smyth (1996)

5 steps: Fayyad et al KDP model:

- 1) *Selection*
- 2) *Preprocessing*
- 3) *Transformation*
- 4) *Data mining*
- 5) *Interpretation & Evaluation*



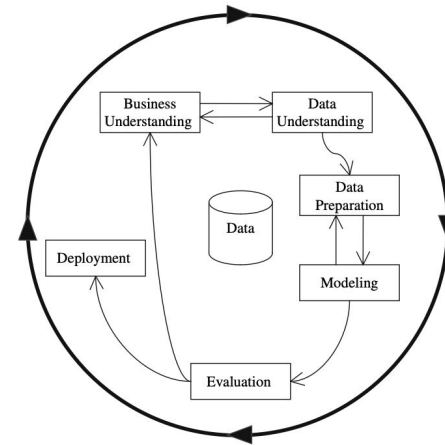


Knowledge Discovery Process (KDP) Models

Industrial Models

- Business understanding
- Data Understanding
- Data preparation
- Modeling
- Evaluation
- Deployment

Data Mining - A Knowledge Discovery Approach by Cis Pedrycz, and Swiniarski



The CRISP-DM KD process model (source: <http://www.crisp-dm.org/>).



QA